

CMOS DIGITAL LOGIC CIRCUITS

Outline

- › Introduction
- › Transistor switch model
- › CMOS implementations of logic gates
- › Digital logic inverters

Introduction

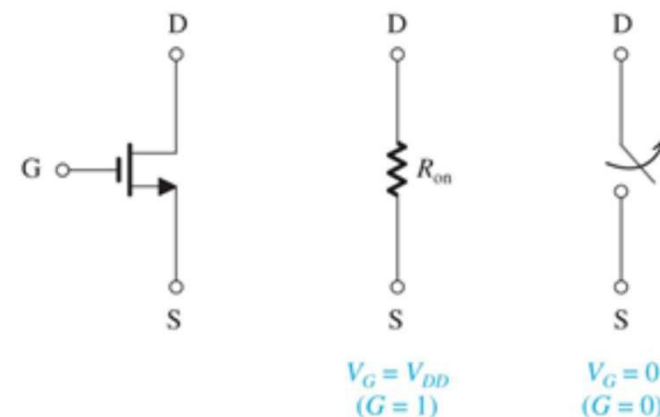
- › First digital devices were Transistor Transistor Logic (TTL) and others made of Bipolar Junction Transistors (BJTs)
- › These were followed by MOS digital devices
- › The initial MOS digital gates were PMOS but these were later replaced by NMOS devices
- › Power dissipation of NMOS microprocessors became an issue and this led to development of **Complimentary MOS (or CMOS)** technology
- › CMOS is by far the most dominant technology in digital device technology

History

- › Digital electronics can be found in many applications in the form of microprocessors, microcontrollers, PCs, DSPs, and an uncountable number of other systems.
- › The design of digital circuits has progressed from resistor-transistor logic (RTL) and diode-transistor logic (DTL) to transistor-transistor logic (TTL) and emitter-coupled logic (ECL) to NMOS and now **complementary MOS (CMOS)**
- › The density and number of transistors in microprocessors has increased from approximately 2300 in the 1971 4-bit 4004 microprocessor to 400 million in more recent P4 chips and are projected to reach ten billion transistors by 2015

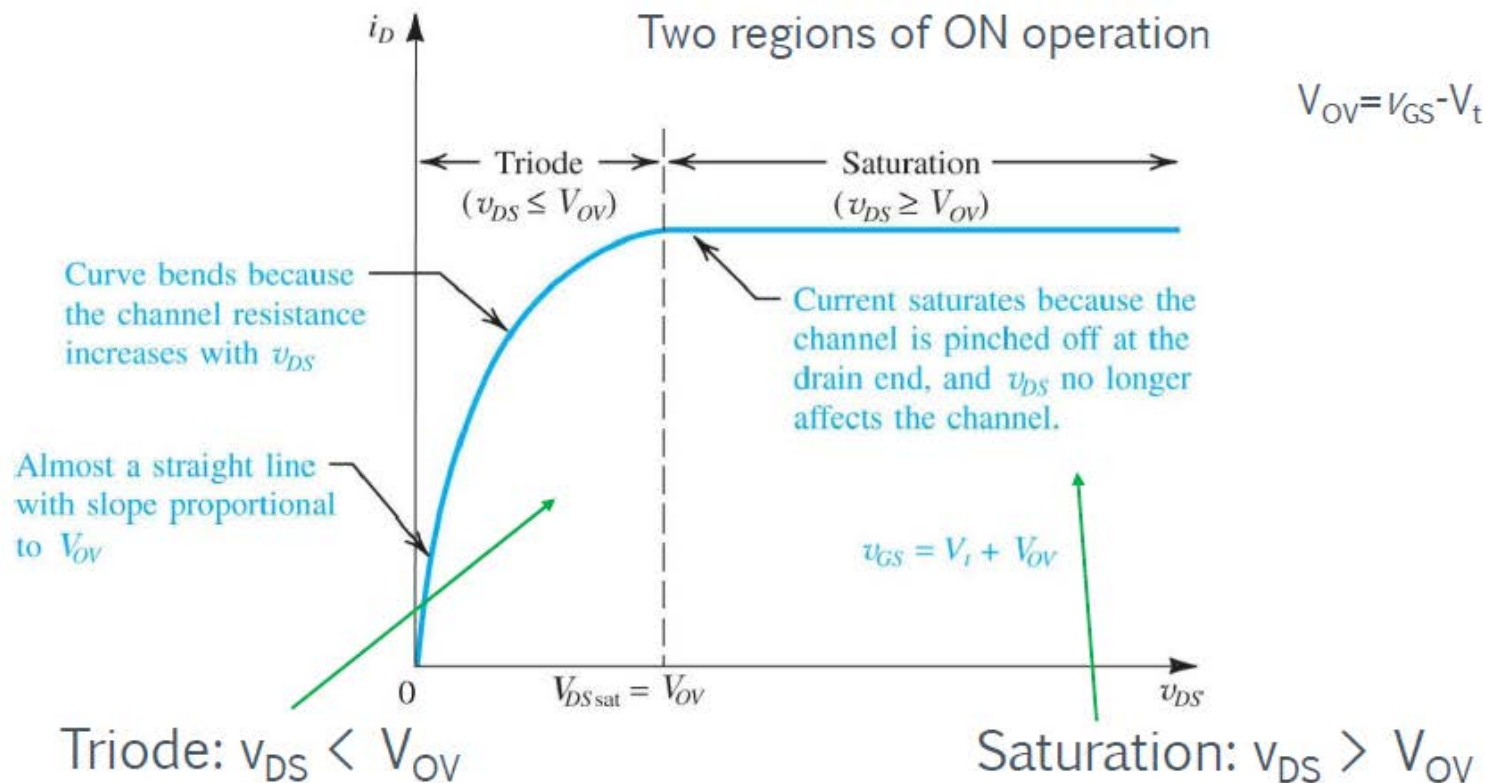
Switch-level transistor model

- › CMOS digital circuits utilize a pair of NMOS and PMOS transistors operating as switches
- › Consider the NMOS transistor switch:
 - › When the gate voltage is high enough the NMOS operates in the triode region
 - › When the gate voltage (or input) is high or logic 1, the switch is closed with a small value of R_{on}
 - › When the gate voltage is zero or low the MOSFET is in cutoff and the switch is open



Operation of the NMOS transistor as an on/off switch

i_D versus v_{DS} for an NMOS

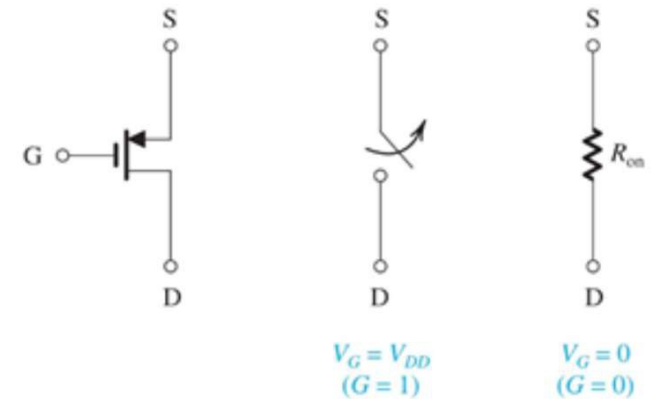


$$i_D = \left[k'_n \left(\frac{W}{L} \right) \left(V_{OV} - \frac{1}{2} v_{DS} \right) \right] v_{DS}$$

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) V_{OV}^2$$

Switch-level transistor model

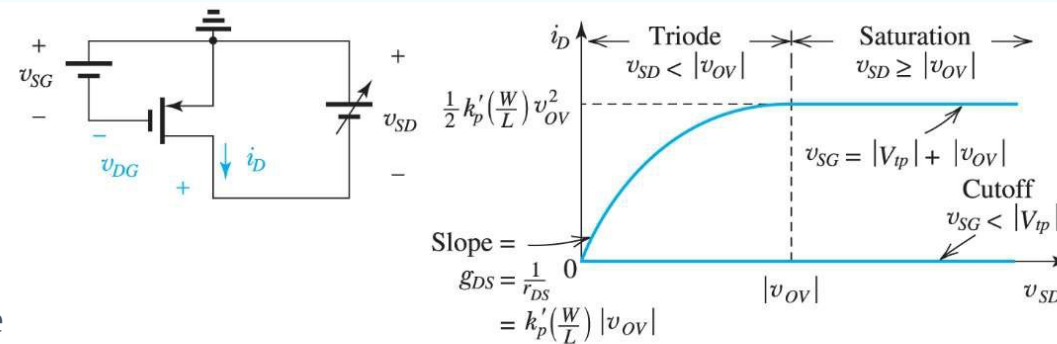
- › For the PMOS transistor switch:
- › When the gate voltage (or input) is high (or V_{DD}) or logic 1 the PMOS is in cutoff and the switch is open
 - V_{SG} has a value that is lower than $|V_{tp}|$
- › When the gate voltage is zero or low the MOSFET is in triode region and the switch is closed with a small value of R_{on}
 - V_{SG} has a value that is higher than $|V_{tp}|$



Operation of the PMOS transistor as an on/off switch

Summary for p-type MOSFET

Table 5.2 Regions of Operation of the Enhancement PMOS Transistor



In **cutoff** region, the MOSFET is off

- $v_{SG} < |V_{tp}|$: no channel; transistor in cutoff; $i_D = 0$
- $v_{SG} = |V_{tp}| + |v_{OV}|$: a channel is induced; transistor operates in the triode region or in the saturation region depending on whether the channel is continuous or pinched off at the drain end;

In **triode** (and cutoff) regions, the MOSFET can be used as a **switch** or a **digital device**

Triode Region

Continuous channel, obtained by:

$$v_{DG} > |V_{tp}|$$

or equivalently

$$v_{SD} < |v_{OV}|$$

Then

$$i_D = k'_p \left(\frac{W}{L} \right) \left[(v_{SG} - |V_{tp}|) v_{SD} - \frac{1}{2} v_{SD}^2 \right]$$

or equivalently

$$i_D = k'_p \left(\frac{W}{L} \right) \left(|v_{OV}| - \frac{1}{2} v_{SD} \right) v_{SD}$$

Saturation Region

Pinched-off channel, obtained by:

$$v_{DG} \leq |V_{tp}|$$

or equivalently

$$v_{SD} \geq |v_{OV}|$$

Then

$$i_D = \frac{1}{2} k'_p \left(\frac{W}{L} \right) (v_{SG} - |V_{tp}|)^2$$

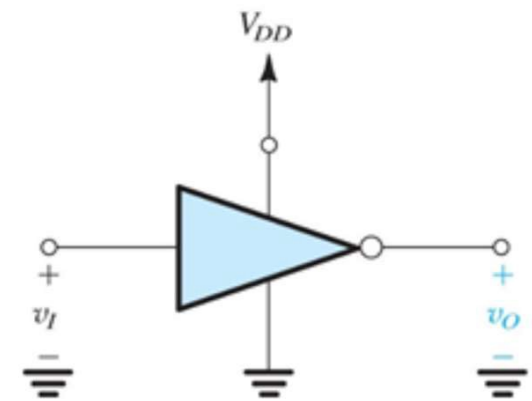
or equivalently

$$i_D = \frac{1}{2} k'_p \left(\frac{W}{L} \right) v_{OV}^2$$

In **saturation** (or “active”) region, the MOSFET can be used as an **amplifier**

Conceptual logic inverter

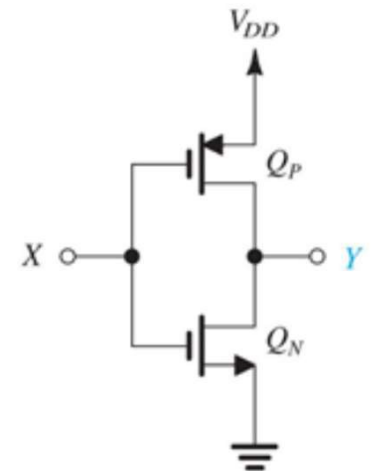
- › An inverter inverts the input
- › When the input is at logic 0, the output will be at logic 1
- › When the input is at logic 1, the output will be at logic 0
- › In terms of actual voltages:
 - When $v_I=0$, $v_o=V_{DD}$
 - When $v_I=V_{DD}$, $v_o=0$



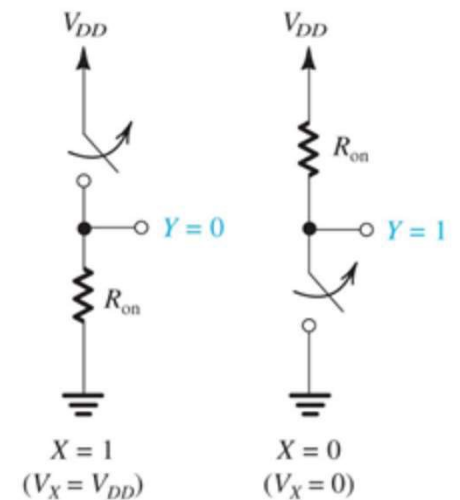
Block diagram of the logic inverter

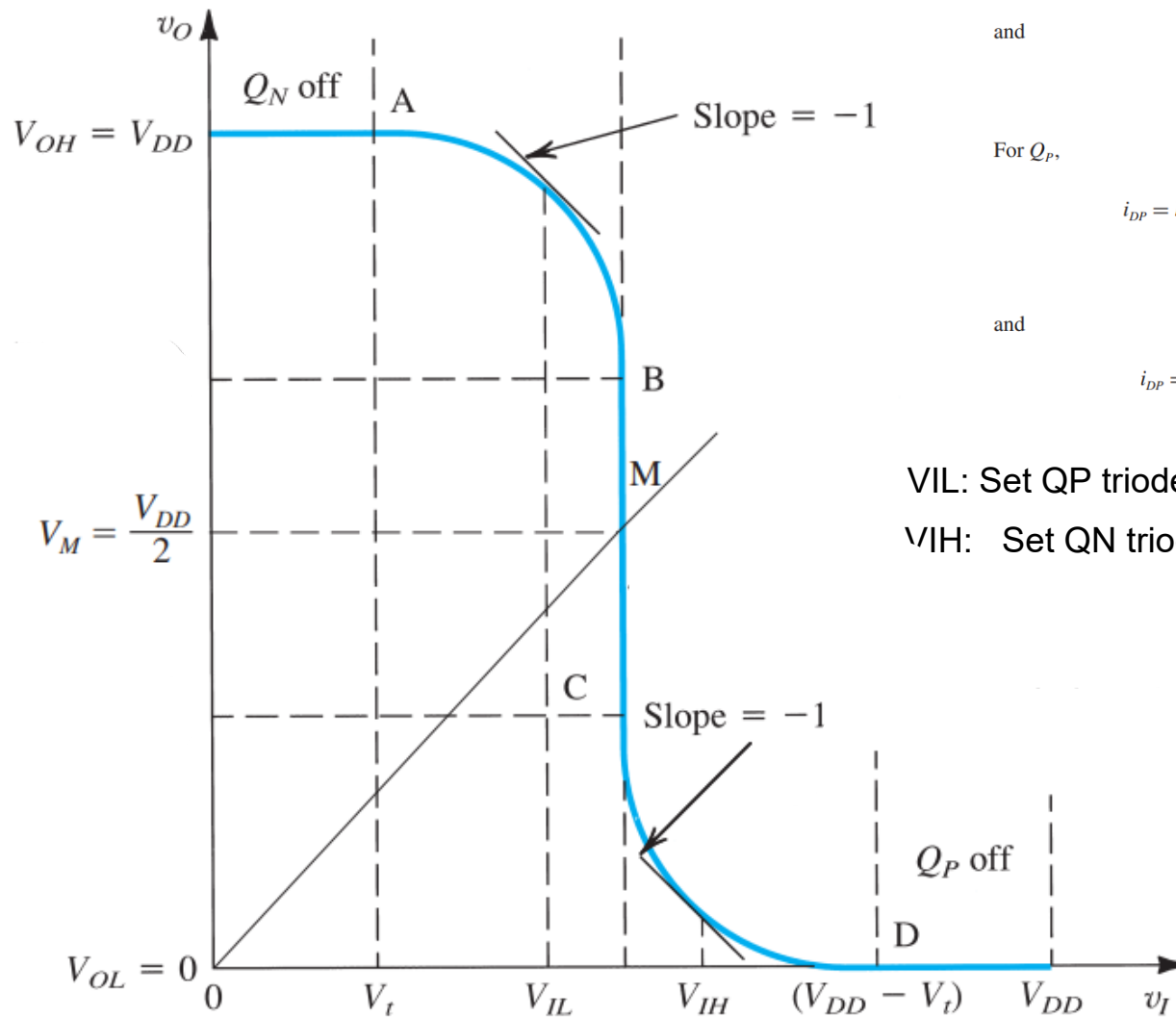
CMOS implementation of an inverter

- › CMOS inverter is shown
- › It consists of an NMOS Q_N and a PMOS Q_P
- › When $X=1$ or $V_X=V_{DD}$, Q_N is on and Q_P is off, $Y=0$ (V_Y is very low)
- › When $X=0$ or $V_X=0$, Q_N is off and Q_P is on, $Y=1$ ($V_Y=V_{DD}$)



CMOS inverter





voltage-transfer curve. For this we need the $i-v$ relationships of Q_N and Q_P . For Q_N ,

$$i_{DN} = k'_n \left(\frac{W}{L} \right)_n \left[(v_I - V_{tn})v_O - \frac{1}{2}v_O^2 \right] \quad \text{for } v_O \leq v_I - V_{tn} \quad (14.28)$$

Vo=Vds; Vgs=Vi

and

$$i_{DN} = \frac{1}{2} k'_n \left(\frac{W}{L} \right)_n (v_I - V_{tn})^2 \quad \text{for } v_O \geq v_I - V_{tn} \quad (14.29)$$

Vds < Vgs-Vtn Triode

For Q_P ,

$$i_{DP} = k'_p \left(\frac{W}{L} \right)_p \left[(V_{DD} - v_I - |V_{tp}|)(V_{DD} - v_O) - \frac{1}{2}(V_{DD} - v_O)^2 \right] \quad \text{for } v_O \geq v_I + |V_{tp}| \quad (14.30)$$

Vds > Vgs-Vtn Saturation

and

$$i_{DP} = \frac{1}{2} k'_p \left(\frac{W}{L} \right)_p (V_{DD} - v_I - |V_{tp}|)^2 \quad \text{for } v_O \leq v_I + |V_{tp}| \quad (14.31)$$

Vsd < Vgs + |Vtp| Triode

Vsd < Vgs + |Vtp| Saturation

VIL: Set QP triode current = QN saturation current

VIH: Set QN triode current = QP saturation current

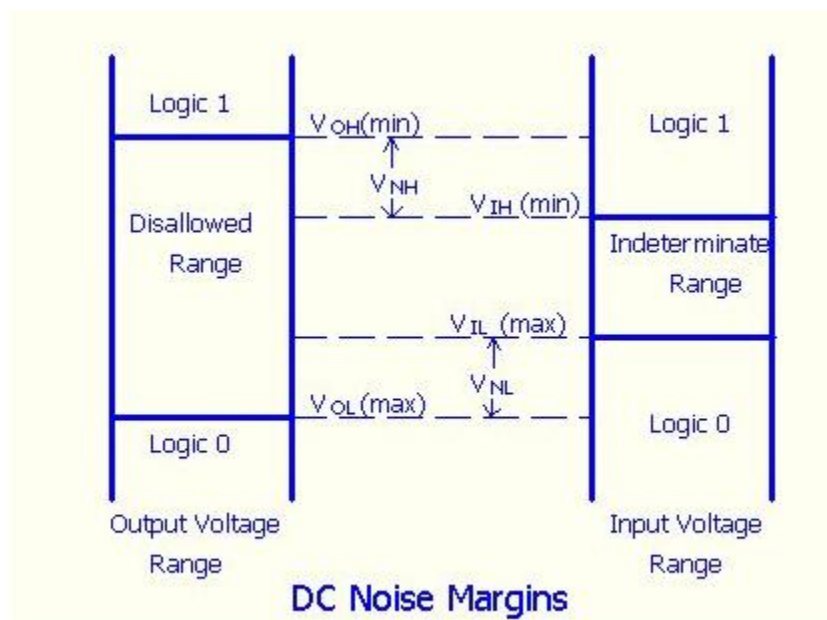
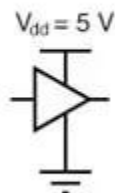
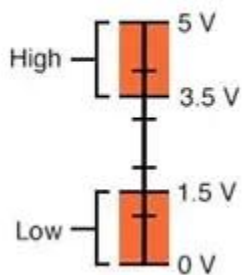
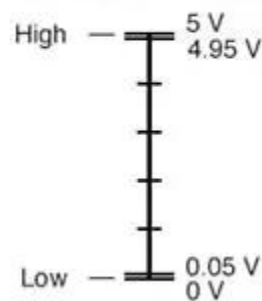


Fig.3: Definition of noise margin

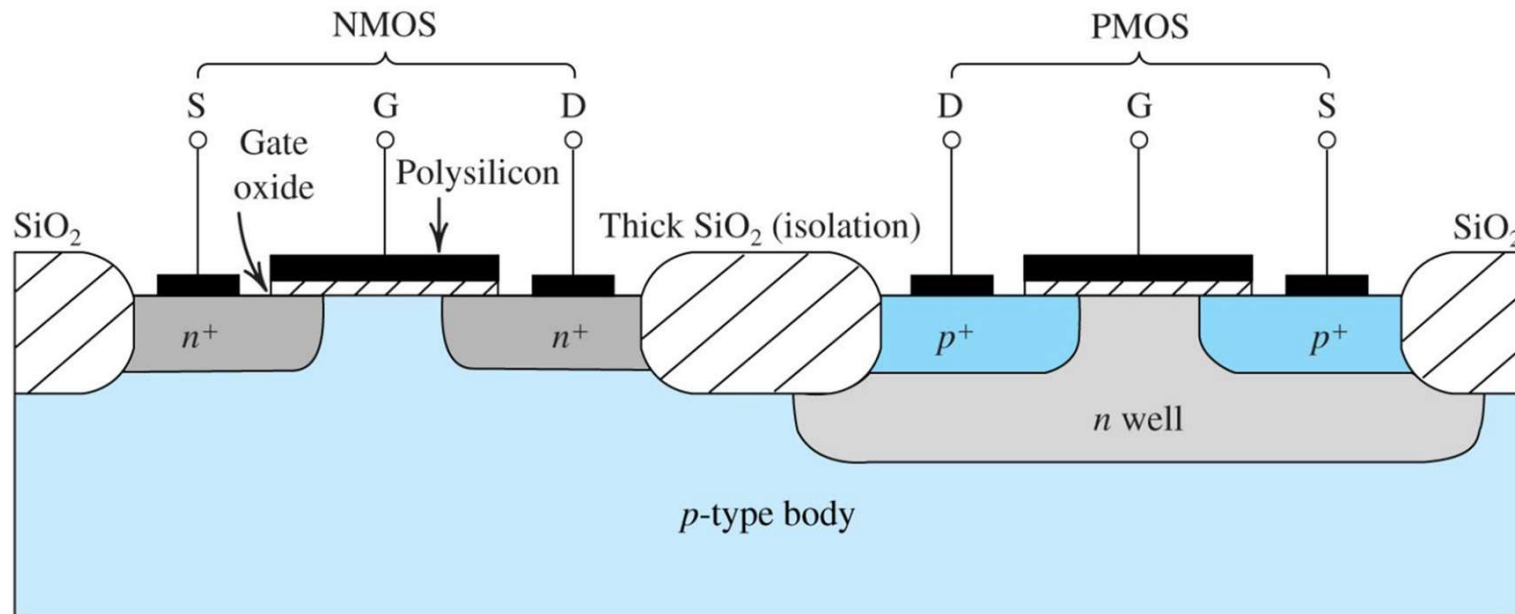
Acceptable CMOS Gate
Input Signal Levels



Acceptable CMOS Gate
Output Signal Levels



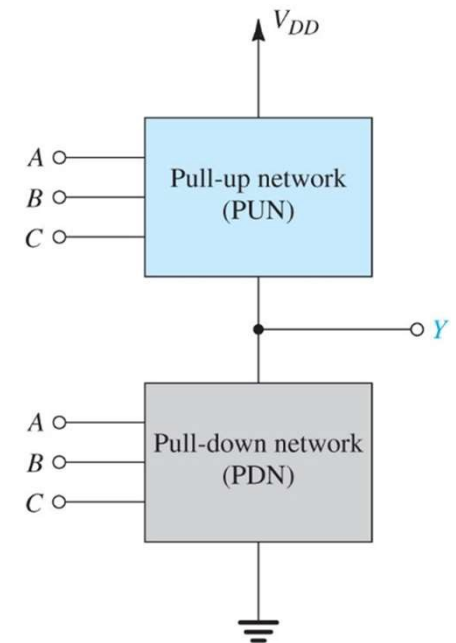
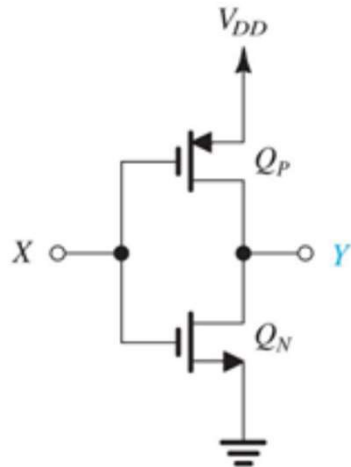
Complimentary MOS or CMOS



- › The NMOS transistor is implemented on a *p*-type substrate
- › The PMOS transistor is fabricated in a specially created *n* region, known as an **n** well
- › The two devices are separated from each other by an oxide layer that acts as an insulator

General structure of CMOS logic gate

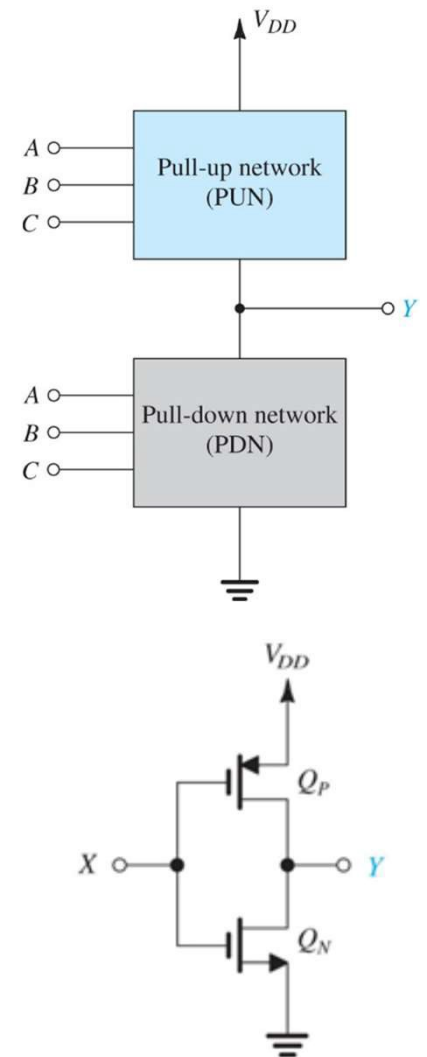
- › The CMOS inverter that we saw had an NMOS pull-down transistor and a PMOS pull-up transistor
- › In general, a CMOS gate consists of
 - A pull-down network with NMOS transistors
 - A pull-up network with PMOS transistors



A three-input
CMOS logic gate

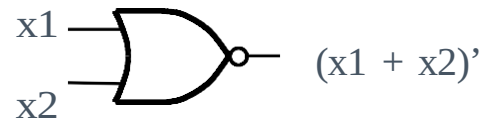
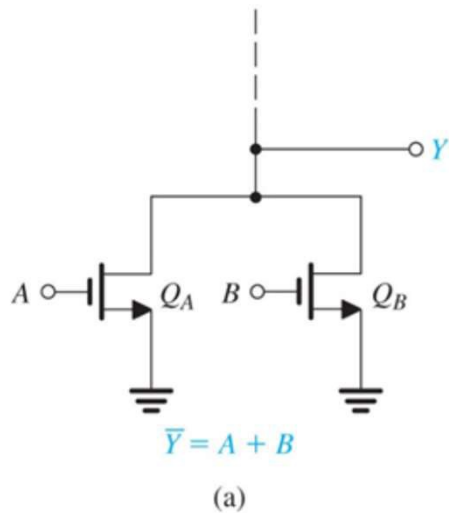
General structure of CMOS logic gate

- › A PDN network is activated when the inputs are high
- › A PUN network is activated when the inputs are low
- › A PDN uses devices in parallel to form an OR function, and devices in series to form an AND function
- › The operation of a PUN is similar to that of a PDN



Examples of pull-down networks (PDN)

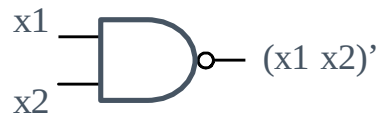
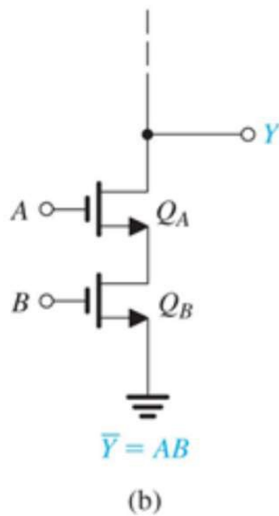
- › Circuit (a) performs a NOT-OR (NOR) function
 - If $A=0$ and $B=0$, $Y=1$ (Y is pulled high)
 - If either A or B (or both) is 1, $Y=0$ (Y is pulled low)



X1	X2	$(X1+X2)'$
0	0	1
0	1	0
1	0	0
1	1	0

Examples of pull-down networks (PDN)

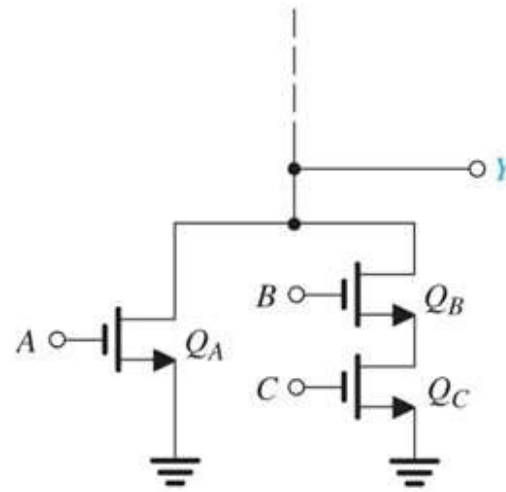
- › Circuit (b) performs a NOT-AND (NAND) function
 - If either A or B (or both) is 0, $Y=1$ (Y is pulled high)
 - If $A=1$ and $B=1$, $Y=0$ (Y is pulled low)



X1	X2	$(X1.X2)'$
0	0	1
0	1	1
1	0	1
1	1	0

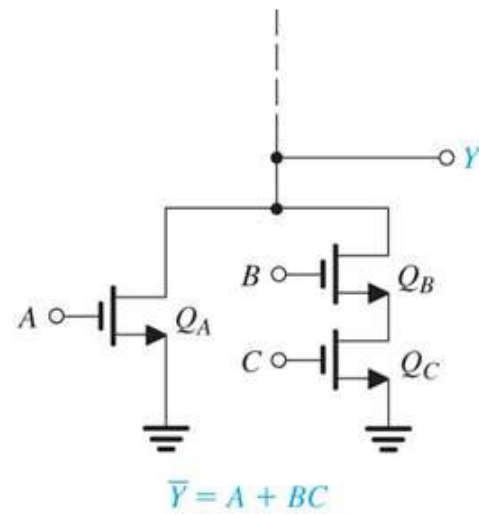
PDN example

- › What is the Boolean function performed by this network?



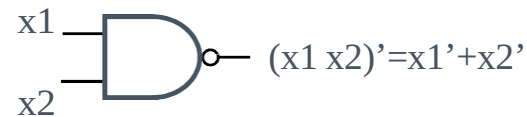
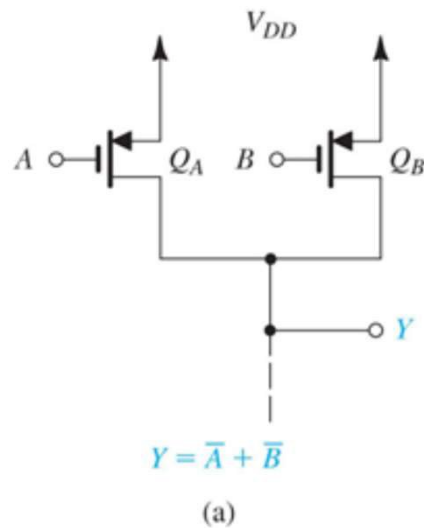
PDN example

- › What is the Boolean function performed by this network?



Examples of pull-up networks (PUN)

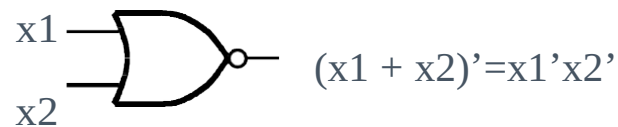
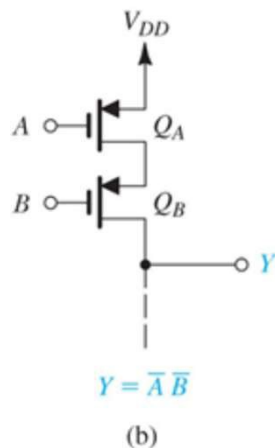
- › Circuit (a) performs a NOT-AND (or NAND) function
 - If either A or B (or both) is 0, Y=1 (Y is pulled high)
 - If A=1 and B=1, Y=0 (Y is pulled low)



X1	X2	(X1'+X2')
0	0	1
0	1	1
1	0	1
1	1	0

Examples of pull-up networks (PUN)

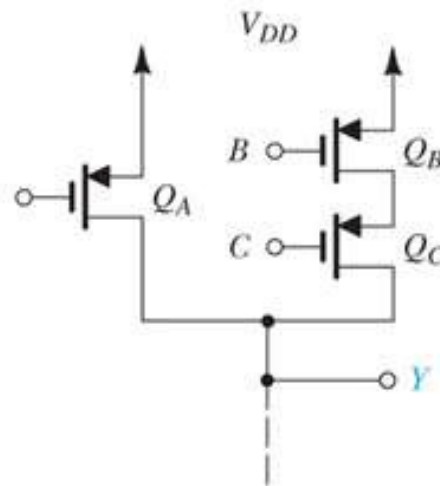
- › Circuit (b) performs a NOT-OR (or NOR) function
 - If either A or B (or both) is 1 Y=0 (Y is pulled low)
 - If A=0 and B=0, Y=1 (Y is pulled high)



X1	X2	(X1'+X2')
0	0	1
0	1	0
1	0	0
1	1	0

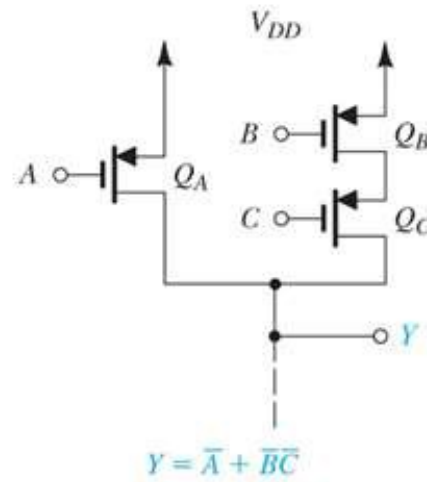
PUN example

- › What is the Boolean function performed by this network?

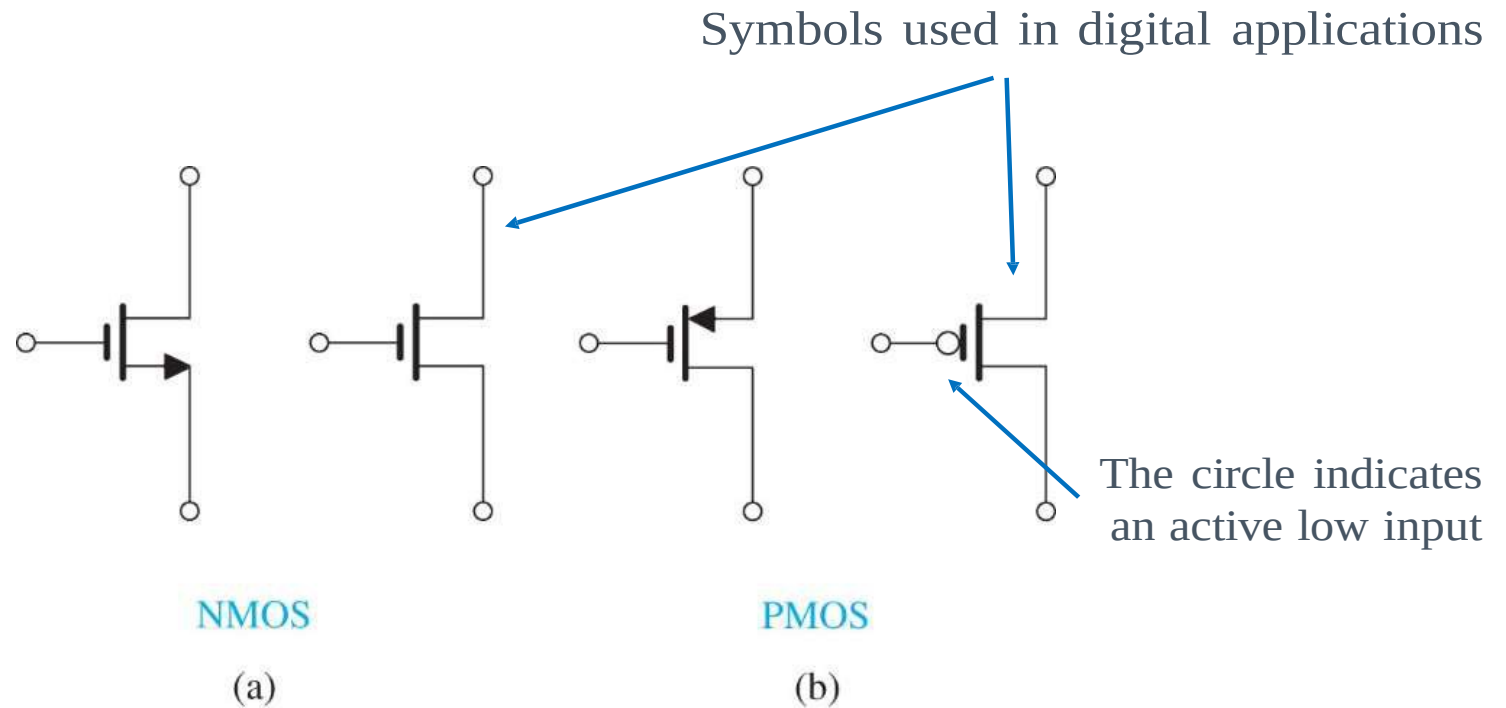


PUN example

- › What is the Boolean function performed by this network?



Alternate MOSFET symbols

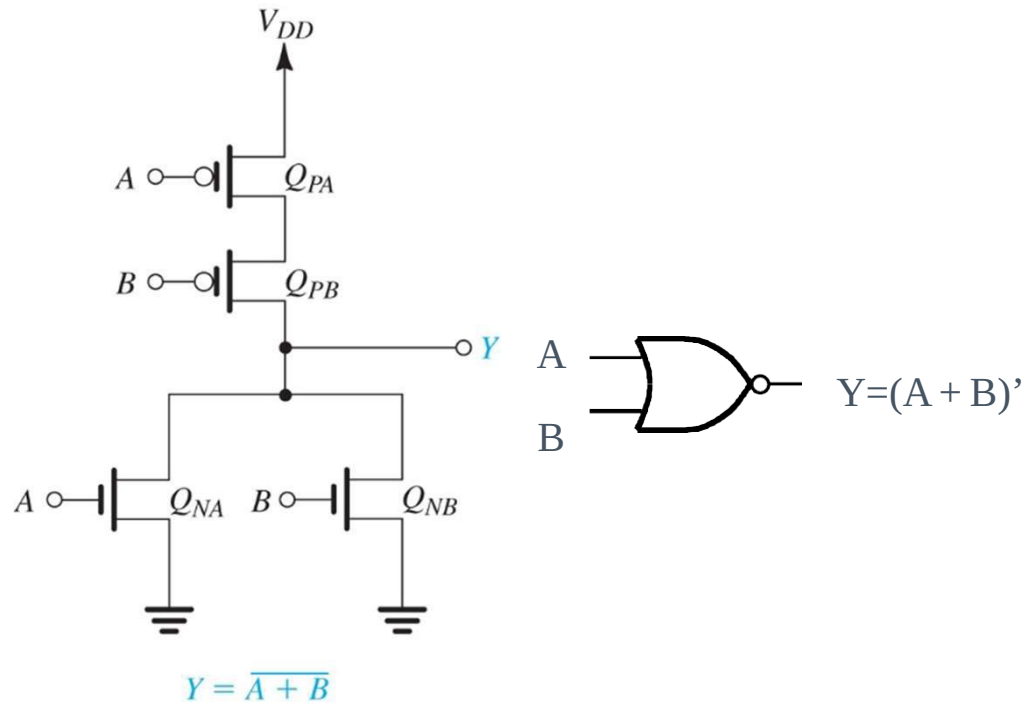


Usual and alternative circuit symbols for MOSFETs.

Complete CMOS NOR gate

CMOS NOR circuit

- If either A or B (or both) is 1, Y=0 (Y is pulled low)
- If A=0 and B=0, Y=1 (Y is pulled high)

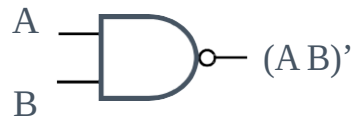
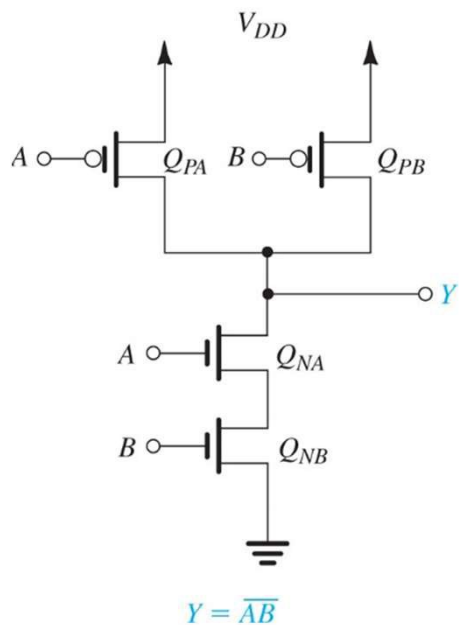


A	B	$(A+B)'$
0	0	1
0	1	0
1	0	0
1	1	0

Complete CMOS NAND gate

CMOS NAND circuit

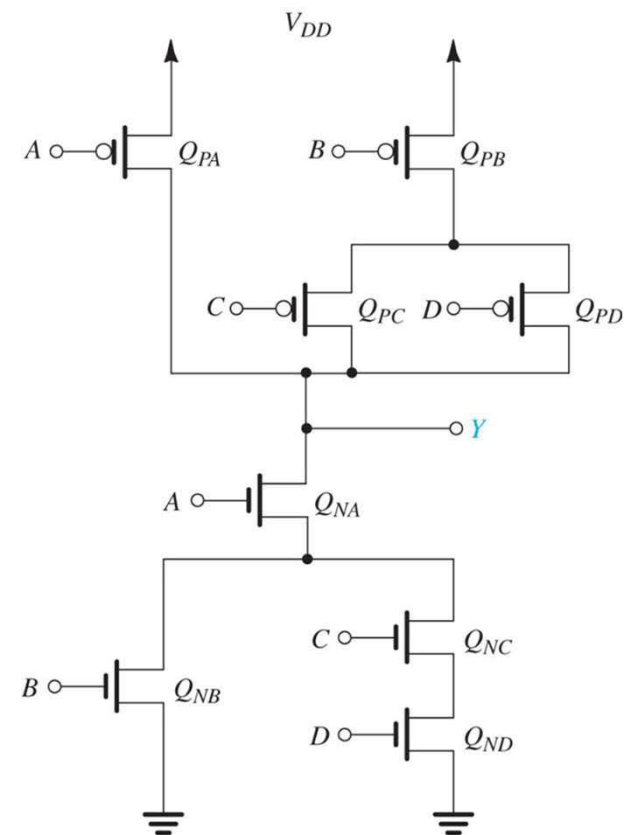
- If $A=1$ and $B=1$, $Y=0$ (Y is pulled low)
- If either A or B (or both) is 0, $Y=1$ (Y is pulled high)



A	B	$(A \cdot B)'$
0	0	1
0	1	1
1	0	1
1	1	0

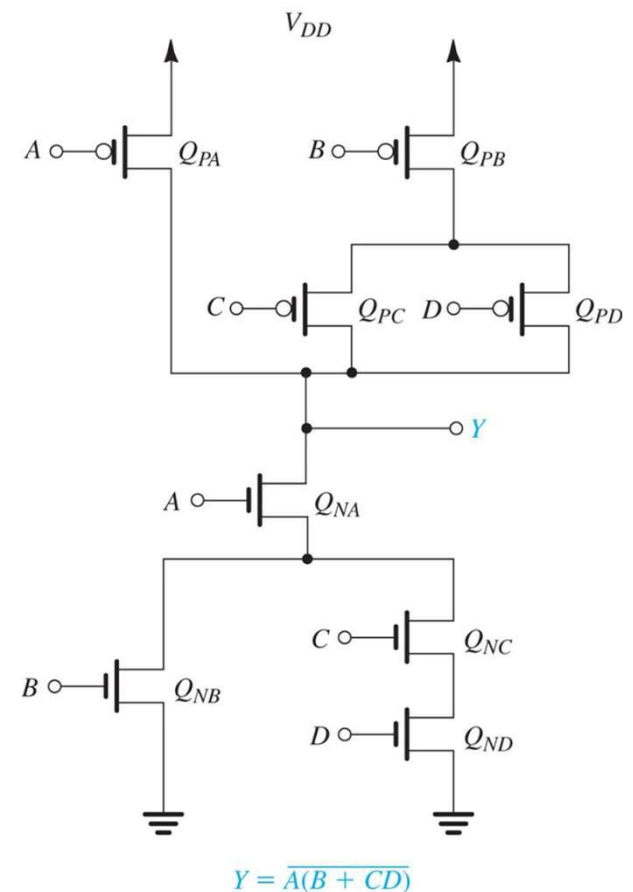
A complex CMOS circuit

- › What is the output Y?



A complex CMOS circuit

- › In the lower half of the circuit,
 - If $A=1$ and ($B=1$ or $C=1$ and $D=1$) then, $Y=0$ (pulled low)
- › In the upper half of the circuit,
 - If $A=0$ or ($B=0$ and ($C=0$ or $D=0$)) then, $Y=1$ (pulled high)
- › $Y = ([A(B+CD)]' + A' + B'(C'+D'))$
- › $Y = A' + B'(C'+D')$



CMOS realization of a complex gate